



EMCH 308 Introduction to Finite Element Stress Analysis

1 Course Information

Course Description

Introduction to stress analysis for beams, plates, shells, and solids using finite element based computer tools.

Pre-requisites: EMCH 260 Solid Mechanics or ENCP 260 with a minimum grade of D;

Co-requisites or Pre-requisites: None.

Required for BSE in Mechanical Engineering

Contact hours - 3

2 Instructor Information

Andrew Gross

Email: andrewgross@sc.edu

Phone: 803-777-0183

Office Location: Horizon I, Room 336

Office Hours: Monday 10:00am-11:30am. Wednesday 12:00pm-1:30pm.

Other: E-mails and Blackboard posts will be replied to within 48 hours.

3 Teaching Assistant/Grader

Francesco Kagugube. yigamawk@email.sc.edu

4 Textbook

The Finite Element Method, ISBN: 9780486411811

5 Meeting Time/Location

Meeting Time

Tuesday & Thursday | 11:40am - 12:55pm

Meeting Location

Swearingen, 1D15

6 Course Topics

1. Introduction to finite element methods and applications
2. Introduction to finite element based computer tools
3. Linear elastic stress analysis for beams
4. Linear elastic stress analysis for plates
5. Linear elastic stress analysis for shells
6. Linear elastic stress analysis for solids

7 Course Outcomes

1. Students will demonstrate the ability to create geometric models for beams, plates, shells, and solids using finite element based computer tools
2. Students will demonstrate the ability to generate finite element meshes based on geometric models for beams, plates, shells, and solids using finite element based computer tools
3. Students will demonstrate the ability to apply boundary conditions to finite element meshes for beams, plates, shells, and solids using finite element based computer tools
4. Students will demonstrate the ability to perform linear elastic stress analysis on finite element models for beams, plates, shells, and solids using finite element based computer tools
5. Students will demonstrate the ability to interpret and communicate finite element stress analysis results for beams, plates, shells, and solids

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Lecture Schedule

Week	Topic
1	Introduction to the finite element method
2	Fundamentals of the finite element method
3	Fundamentals of the finite element method
4	Basics of coding in python and introduction to Abaqus
5	Trusses and beams
6	Plane stress and plane strain (2D problems)
7	Plane stress and plane strain (2D problems)
8	Axisymmetric problems and contact
9	Solid elements
10	Metal plasticity
11	Shell Elements
12	Buckling and postbuckling
13	Explicit time integration
14	Verification and Validation

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Grading Policy

General Assignment Information

All homework exercises are available to students from the beginning of the semester. Exercises can be turned in whenever completed, but no later than the due date. Due dates will be provided in class at least three days before an exercise becomes due. Submission of solutions to Blackboard must be made prior to 5:00 pm on the due date.

Exams

There will be one midterm exam and it will be conducted in class via Blackboard. The midterm exam will be 75 minutes in length and will occur during class time. Any student who does not take the exam at this time must have prior approval from the instructor or documentation of an extenuating circumstance.

Final Project

Students will work in small groups on a semester long project. Each group will develop independent

solutions to the same design challenge. Each group will work with a machining student at Midlands Tech to fabricate their design. A visit to Midlands Tech will occur during class time, likely in September. Several project checkpoints will be due throughout the semester and a final report will be due on

Tuesday, December 8th at 11:59 pm.

Evaluation and Grading Scale

All grades will be posted on Blackboard. You are strongly encouraged to check your scores in Blackboard regularly. A final letter grade will be assigned based on percentages.

Assignment Weights

Homework	40%
Class Participation	15%
Midterm Exam	20%
Semester Project	25%

Grading Scale

A	89.5% - 100%
B+	84.5% - 89.4%
B	79.5% - 84.4%
C+	74.5% - 79.4%
C	69.5% - 74.4%
D+	64.5% - 69.4%
D	59.5% - 64.4%
F	0% - 59.4%

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Attendance Policy

When you miss class, you miss important information. If you are absent, you are responsible for learning material covered in class. If you have an [excused absence](https://academicbulletins.sc.edu/undergraduate/policies-regulations/undergraduate-academic-regulations) (<https://academicbulletins.sc.edu/undergraduate/policies-regulations/undergraduate-academic-regulations>), you will be permitted to make up coursework.

To arrange excuses for absences that can be anticipated at the start of the term, you should: *i*) Submit a request in writing (email is acceptable) stating the dates of the anticipated absence no later than the end of the second week of the course. *ii*) Explain the reason for absence. In some cases, documentation may be required. Please [consult the policy](https://academicbulletins.sc.edu/undergraduate/policies-regulations/undergraduate-academic-regulations) (<https://academicbulletins.sc.edu/undergraduate/policies-regulations/undergraduate-academic-regulations>) for additional information. *iii*) Include any request for make-up work.

To arrange excuses for absences that cannot be anticipated at the start of the term, (e.g. legal proceedings or illness), you should, at the first opportunity, submit in writing a request stating: *i*) The date of absence. *ii*) The reason for absence. In some cases, documentation may be required. Please [consult the policy](https://academicbulletins.sc.edu/undergraduate/policies-regulations/undergraduate-academic-regulations) for additional information. *iii*) Any request for make-up work as soon as reasonably possible after you become aware of the need to be absent.

Class participation requires in class attendance, abstention from unrelated content during class meetings, completion of required readings, watching of instructional videos prior to class, and progress on assignments. Participation will be assessed at every class meeting.

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Miscellaneous Policies

Academic Integrity

University policies and procedures regarding academic integrity are defined in policy STAF 6.25, Academic Responsibility - The Honor Code (see <http://www.sc.edu/policies/ppm/staf625.pdf>). Prohibited behaviors include plagiarism, cheating, falsification, and complicity. All potential Honor Code violations will be reported to the Office of Academic Integrity, which has the authority to implement non-academic penalties as described in STAF 6.25. Academic penalties for Honor Code violations in this course range from a zero on the assignment to failure of the course.

Diversity

In order to learn, we must be open to the views of people different than ourselves. In this time we share together over the semester, please honor the uniqueness of your fellow classmates and appreciate the opportunity we have to learn from one another. Please respect each others' opinions and refrain from personal attacks or demeaning comments of any kind. Finally, remember to keep confidential all issues of a personal or professional nature that are discussed in class.

Accommodating Disability

Reasonable accommodations are available for students with a documented disability. If you have a disability and may need accommodations to fully participate in this class, contact the Student Disability Resource Center: 777-6142 email sasds@mailbox.sc.edu, or stop by 1705 College Street, Close-Hipp, Suite 102, Columbia, SC 29208.

All accommodations must be approved through the Student Disability Resource Center.

Technical Support

Students facing issues with Blackboard or Microsoft Teams can submit requests on <https://scprod.service-now.com/sp> or call 803-777-1800. Also, departmental support is available on helpme@cec.sc.edu.

Interactions

S2I: Blackboard announcements will be used with a combination of emails/teams for further interaction.

S2S: Communication between students is required for the semester project. Each group may use their preferred means.

S2C: All course material is posted on Blackboard.

Online Learning Outcomes

All Learning Outcomes are equivalent of those of a face-to-face (F2F) version of the course.

Fair Policy

All readings/materials comply with copyright/fair use policies.

Technology

Course delivery will be conducted through Blackboard Ultra/Microsoft Teams.

Course assignments will make use of the commercial finite element software Abaqus. Abaqus is available remotely via <https://view.cec.sc.edu/> and in person at the CEC computer labs. Instructions and a demo on how to remotely access Abaqus will be provided in class.

SO/CO Table	CO 1	CO 2	CO 3	CO 4	CO 5
SO 1 - Complex Problems	0	0	0	0	0
SO 2 - Design	0	0	0	0	0
SO 3 - Communication	0	0	0	0	0
SO 4 - Judgment	0	0	0	0	0
SO 5 - Teams	0	0	0	0	0
SO 6 - Experiments	0	0	0	0	0
SO 7 - Knowledge	0	0	0	0	0

SO 1 - Complex Problems

an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics

SO 2 - Design

an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors

SO 3 - Communication

an ability to communicate effectively with a range of audiences

SO 4 - Judgment

an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts

SO 5 - Teams

an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives

SO 6 - Experiments

an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions

SO 7 - Knowledge

an ability to acquire and apply new knowledge as needed, using appropriate learning strategies.

This course assumes that work submitted by students for the midterm exam and exercises 1-12 – will be generated by the students themselves, working individually or in groups as directed by class assignment instructions. This policy indicates the following constitute violations of academic honesty: a student has another person/entity do the work of any substantive portion of a graded assignment for them, which includes purchasing work from a company, hiring a person or company to complete an assignment or exam, and/or using generative AI tools (such as ChatGPT).

The responsible use of AI tools is allowed and encouraged for the semester project and exercises 13-22, except for the .jnl file produced by using the Abaqus software. AI modified or generated .jnl files and AI produced images of finite element method results constitute violations of academic honesty.